

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from google.colab import files
uploaded = files.upload()

<IPython.core.display.HTML object>

Saving Amazon Sale Report.csv to Amazon Sale Report.csv

df = pd.read_csv("Amazon Sale Report.csv")
df.head()

{"type": "dataframe", "variable_name": "df"}

df.shape

(128976, 19)

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 128976 entries, 0 to 128975
Data columns (total 19 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   index                                128976 non-null  int64
1   Order ID                             128976 non-null  object
2   Date                                 128976 non-null  object
3   Status                               128976 non-null  object
4   Fulfilment                           128976 non-null  object
5   Sales Channel                        128976 non-null  object
6   ship-service-level                   128976 non-null  object
7   Category                             128976 non-null  object
8   Size                                 128976 non-null  object
9   Courier Status                       128976 non-null  object
10  Qty                                  128976 non-null  int64
11  currency                             121176 non-null  object
12  Amount                               121176 non-null  float64
13  ship-city                            128941 non-null  object
14  ship-state                           128941 non-null  object
15  ship-postal-code                     128941 non-null  float64
16  ship-country                         128941 non-null  object
17  B2B                                  128976 non-null  bool
18  fulfilled-by                         39263 non-null  object
dtypes: bool(1), float64(2), int64(2), object(14)
memory usage: 17.8+ MB

df.dropna(subset=["Amount"])

```

```
{"type": "dataframe"}
```

```
df.info()
```

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<class 'pandas.core.frame.DataFrame'>
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memory usage: 17.8+ MB
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```
df.shape
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```
(128976, 19)
```

```
plt.figure(figsize=(8,5))
```

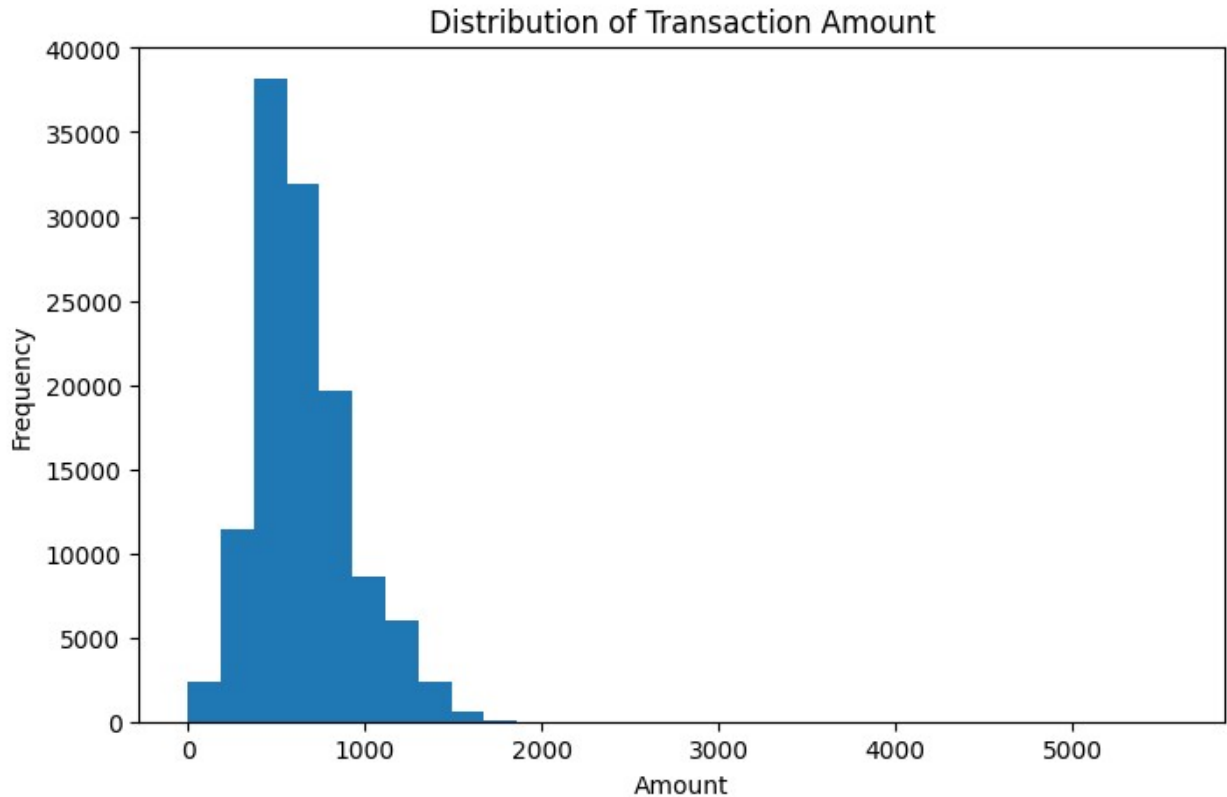
```
plt.hist(df["Amount"], bins=30)
```

```
plt.title("Distribution of Transaction Amount")
```

```
plt.xlabel("Amount")
```

```
plt.ylabel("Frequency")
```

```
plt.show()
```



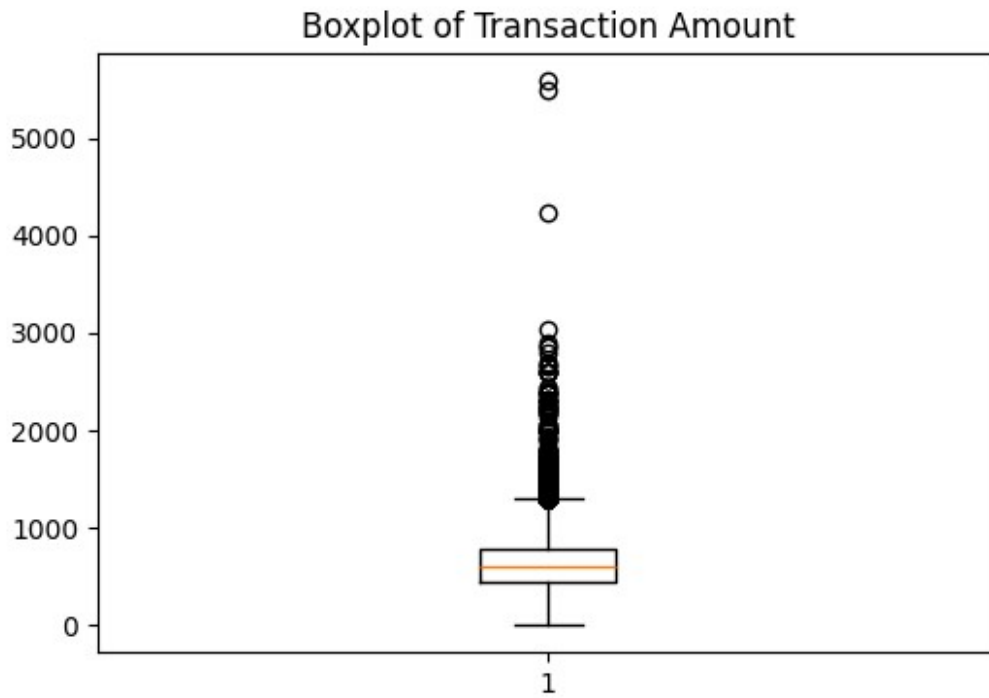
The histogram shows a right-skewed distribution of transaction amounts. Most purchases are concentrated in the lower range, particularly between ₹500–₹1000, indicating that customers primarily buy low-to-mid priced products. High-value transactions are comparatively rare.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("Amazon Sale Report.csv")
df["Amount"].dtype
dtype('float64')

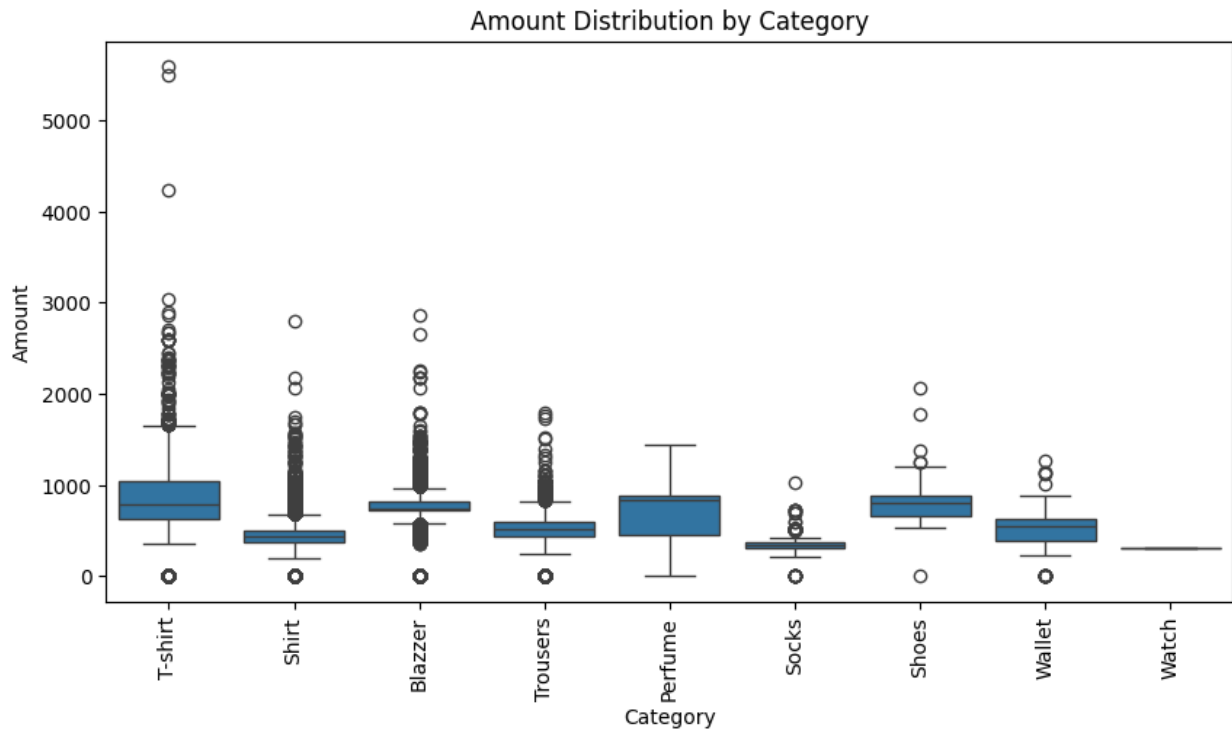
df["Amount"] = pd.to_numeric(df["Amount"], errors="coerce")
df = df.dropna(subset=["Amount"])

plt.figure(figsize=(6,4))
plt.boxplot(df["Amount"])
plt.title("Boxplot of Transaction Amount")
plt.show()
```



The boxplot highlights the presence of numerous high-value outliers in the Amount column. While the majority of transactions fall within a moderate range, several extreme values indicate unusually large purchases.

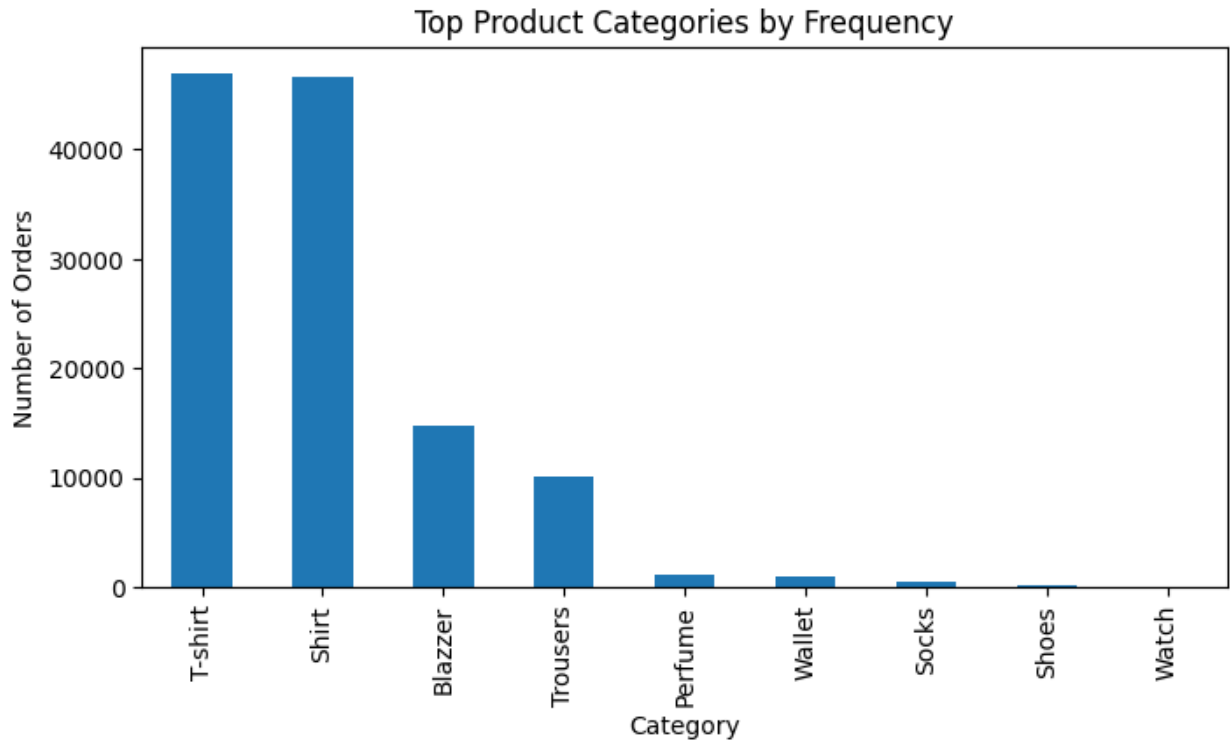
```
plt.figure(figsize=(10,5))
sns.boxplot(x="Category", y="Amount", data=df)
plt.xticks(rotation=90)
plt.title("Amount Distribution by Category")
plt.show()
```



The category-wise boxplot reveals significant differences in transaction amounts across product types. Categories like T-shirts and Blazers display wider spreads and higher outliers, indicating greater pricing variability. In contrast, categories such as Socks and Watches exhibit relatively stable price distributions.

```
category_counts = df["Category"].value_counts()

plt.figure(figsize=(8,4))
category_counts.plot(kind="bar")
plt.title("Top Product Categories by Frequency")
plt.xlabel("Category")
plt.ylabel("Number of Orders")
plt.show()
```



The bar chart reveals that product sales are highly concentrated within a few categories. T-shirts and Shirts dominate transaction frequency, while categories such as Watches, Shoes, and Socks contribute significantly fewer orders. This indicates an uneven category distribution in customer purchasing behavior.

The exploratory analysis of Amazon sales data reveals that customer purchases are predominantly concentrated in the lower transaction range, indicating a preference for affordable products. The distribution of transaction amounts is right-skewed, with only a limited number of high-value purchases. Category-wise analysis shows noticeable pricing variability across product types, while frequency analysis highlights that only a few categories dominate sales volume. Overall, the data reflects typical retail behavior with strong category concentration and occasional high-value outliers.

KEY PROBLEMS IDENTIFIED:

High Frequency vs High Value Disparity Certain categories such as T-shirts and Shirts dominate transaction volume but may not proportionally contribute to overall revenue due to lower individual transaction amounts.

Presence of High-Value Outliers Several unusually large transaction amounts are observed, which may distort average-based analysis and require special handling or segmentation.

Uneven Category Distribution Product sales are heavily skewed toward a limited number of categories, suggesting portfolio imbalance and potential dependency risks.

Pricing Variability Within Categories Some categories display wide spreads in transaction amounts, indicating inconsistent pricing structures or diverse product mixes.

Recommendations:

Implement category-wise performance monitoring dashboards

Investigate and manage high-value outliers separately

Apply standardized pricing strategies within variable categories

Promote underperforming product categories for balanced sales distribution